Summer Project

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Abstract

This project develops a simulation framework for modelling and compensating thermally induced drift in a thermo-optical medium. The optical setup consists of a Gaussian input field evaluated at the spatial light modulator (SLM) plane, where a phase-only modulation pattern is applied. The modulated field then propagates through the medium, described by a time-varying complex transmission operator, and intensity is recorded at a downstream detector—the sole accessible output of the system.

The thermal model treats the medium as a heat-conducting body governed by the heat equation, with material parameters density, specific heat, and thermal conductivity. Temperature variations modify the refractive index via the thermo-optic effect, introducing a spatially and temporally varying phase perturbation. At the system level this manifests as a slow drift in the effective optical transmission operator, which degrades the response seen at the detector if left uncompensated.

To capture this thermal dynamics without relying on full numerical discretisation at every time step, a physics-informed neural network (PINN) is employed as a surrogate model. The network is trained to satisfy the heat equation at interior collocation points while remaining consistent with available boundary and initial conditions, yielding a physics-constrained approximation of the temperature field. The predicted temperature field is then linked to the optical model through the thermo-optic relation to track drift in the transmission operator over time.

The control objective is to select SLM phase patterns that maximise intensity at a designated target location, accounting for the time-varying state of the thermal medium. This is primarily a modelling and simulation project, combining optical physics, thermal modelling, and physics-informed learning.

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1 Background Information

1.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis (z-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{1}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{2}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{3}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{4}$$

Here w_0 is the beam waist, $w(z)$ is the beam radius at position z, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and z_R is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{5}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and A absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{6}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix T, and the detector measures only intensity:

$$E_{\text{out}} = T\,E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{7}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

1.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{8}$$

where ρ is the density, c_p is the specific heat capacity, κ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{9}$$

where n_0 is the reference refractive index at temperature T_0, and dn/dT is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{10}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{11}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

1.3 SLM Behaviour

The SLM acts only in the transverse (x, y) plane and is modeled as a phase-only modulator. The applied phase pattern $\phi(x, y)$ is the control input and sets the relative phases across the aperture, thereby altering interference at the output. The modulated field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

1.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) are a class of machine-learning models designed to approximate the solution of a physical system while explicitly respecting its governing equations. In a standard neural network, one typically provides input–output data pairs and trains the network to reproduce the observed mapping. A PINN extends this idea by requiring the network not only to fit data, but also to satisfy the differential equations, boundary conditions, and initial conditions that describe the underlying physics.

In the present project, the quantity of interest is the temperature field inside the thermo-optical material. Rather than solving for this field using only a conventional numerical discretisation, a PINN represents the temperature by a neural network

$$T_\theta(\mathbf{r}, t), \tag{12}$$

where $\mathbf{r}$ denotes spatial position, t denotes time, and θ represents the trainable parameters of the network, namely its weights and biases. The role of training is to adjust θ so that the network output approximates a temperature field that is both physically admissible and consistent with any available observations.

For the thermal model considered here, the governing equation is the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t). \tag{13}$$

A PINN does not enforce this equation symbolically; instead, it enforces it approximately by penalising violations of the equation at many sampled points in space and time. Substituting the network output $T_\theta(\mathbf{r}, t)$ into the heat equation gives the residual

$$\mathcal{R}_\theta(\mathbf{r}, t) = \rho c_p \frac{\partial T_\theta}{\partial t} - \nabla \cdot (\kappa \nabla T_\theta) - Q(\mathbf{r}, t). \tag{14}$$

If T_θ were the exact physical solution, this residual would vanish everywhere. The PINN is therefore trained so that $\mathcal{R}_\theta(\mathbf{r}, t)$ is as small as possible over the domain of interest.

The overall training objective is written as a weighted sum of loss terms. To distinguish the different point sets used in training, let $\{(\mathbf{r}_i^{(f)}, t_i^{(f)})\}_{i=1}^{N_f}$ denote the interior collocation points used to enforce the PDE, $\{(\mathbf{r}_i^{(d)}, t_i^{(d)}, T_i^{\text{data}})\}_{i=1}^{N_d}$ the observed data points, $\{(\mathbf{r}_i^{(b)}, t_i^{(b)}, T_i^{\text{BC}})\}_{i=1}^{N_b}$ the boundary-condition samples, and $\{\mathbf{r}_i^{(0)}\}_{i=1}^{N_0}$ the spatial points at the initial time $t = 0$, with prescribed initial temperature $T^{\text{init}}(\mathbf{r})$. The component losses are then

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| \mathcal{R}_\theta(\mathbf{r}_i^{(f)}, t_i^{(f)}) \right|^2, \tag{15}$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \left| T_\theta(\mathbf{r}_i^{(d)}, t_i^{(d)}) - T_i^{\text{data}} \right|^2, \tag{16}$$

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| T_\theta(\mathbf{r}_i^{(b)}, t_i^{(b)}) - T_i^{\text{BC}} \right|^2, \tag{17}$$

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \left| T_\theta(\mathbf{r}_i^{(0)}, 0) - T^{\text{init}}(\mathbf{r}_i^{(0)}) \right|^2. \tag{18}$$

The total loss is

$$\mathcal{L} = \lambda_{\text{data}}\mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}}. \tag{19}$$

Here N_f, N_d, N_b, and N_0 are the numbers of PDE, data, boundary, and initial samples, respectively. The quantities T_i^{data} and T_i^{BC} are the prescribed temperature values at the data and boundary sample points, while $T^{\text{init}}(\mathbf{r}_i^{(0)})$ is the prescribed initial temperature at spatial point $\mathbf{r}_i^{(0)}$. The coefficients λ_{data}, λ_{PDE}, λ_{BC}, and λ_{IC} are scalar weighting hyperparameters that determine the relative contribution of each loss term during training.

In training, the network takes $(\mathbf{r}, t)$ as input and outputs $T_\theta(\mathbf{r}, t)$. Automatic differentiation provides the required spatial and temporal derivatives in $\mathcal{R}_\theta$, and the parameters θ are updated to minimise $\mathcal{L}$, yielding a temperature field that balances data fidelity with physical consistency.

Once a thermally consistent approximation $T_\theta(\mathbf{r}, t)$ has been obtained, it can be coupled to the optical model through the thermo-optic relation

$$n_\theta(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T_\theta(\mathbf{r}, t) - T_0), \tag{20}$$

which gives a corresponding refractive-index field. This in turn induces a time-dependent optical phase perturbation,

$$\Delta\phi_\theta(x, y, t) = k \int \left(n_\theta(x, y, z, t) - n_0\right) dz, \tag{21}$$

and therefore modifies the effective optical response of the material. In the present setting, the PINN is thus used as a physics-constrained surrogate model for the thermal dynamics that drive slow drift in the optical system.

The main appeal of this approach is that it incorporates known thermal physics directly into the learning process. As a result, the model does not rely solely on large labelled datasets, and its predictions are encouraged to remain consistent with the governing equations of the system. This makes PINNs a natural candidate for modelling thermally driven changes in an optical medium whose internal state is only partially observed through optical measurements.

1.5 How would one solve the 1D heat equation with a PINN?

To make the PINN methodology concrete, consider the one-dimensional heat equation

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + q(x,t),$$

defined on a spatial domain $x \in [0, L]$ and for $t > 0$, where $\alpha > 0$ is the thermal diffusivity and $q(x,t)$ is a prescribed heat source. The problem is closed by an initial condition

$$T(x,0) = f(x)$$

and Dirichlet boundary conditions

$$T(0,t) = g_0(t), \qquad T(L,t) = g_L(t).$$

The goal is to find a function $T(x,t)$ satisfying all three of these requirements simultaneously.

The neural network as a function approximator

A PINN replaces the unknown solution $T(x,t)$ with the output of a feedforward neural network $T_\theta(x,t)$. The network takes a single space-time point as its two-dimensional input and produces a scalar temperature prediction:

$$a^{[0]} = \begin{bmatrix} x \\ t \end{bmatrix}, \qquad a^{[N]} = T_\theta(x,t).$$

The forward pass through the network proceeds layer by layer. For $\ell = 1, \ldots, N$, a pre-activation vector is formed by a linear transformation of the previous layer's output,

$$z^{[\ell]} = W^{[\ell]} a^{[\ell-1]} + b^{[\ell]},$$

and a nonlinear activation function σ is then applied elementwise to produce the next layer's output,

$$a^{[\ell]} = \sigma\left(z^{[\ell]}\right), \qquad \ell = 1, \ldots, N-1.$$

Here $W^{[\ell]}$ is the weight matrix for layer ℓ, $b^{[\ell]}$ is the corresponding bias vector, $z^{[\ell]}$ is the pre-activation at layer ℓ, and $a^{[\ell]}$ is its post-activation output. The final layer $a^{[N]}$ is a single scalar and is interpreted directly as the predicted temperature $T_\theta(x,t)$; no activation function is applied at the output.

Encoding the physics via automatic differentiation

The network itself knows nothing about the heat equation. Physics enters through the loss function. Because $T_\theta(x,t)$ is a differentiable composition of smooth functions, its partial derivatives with respect to the inputs can be computed exactly by automatic differentiation:

$$T_t = \frac{\partial T_\theta}{\partial t}, \qquad T_{xx} = \frac{\partial^2 T_\theta}{\partial x^2}.$$

Substituting the network output into the heat equation yields the PDE residual

$$R_\theta(x,t) = \frac{\partial T_\theta}{\partial t} - \alpha \frac{\partial^2 T_\theta}{\partial x^2} - q(x,t).$$

If T_θ were the true solution, this residual would vanish everywhere in the domain. Training is designed to make it as small as possible.

The loss function

The training objective is a weighted sum of three mean-squared loss terms, one for each requirement imposed on the solution.

The *PDE residual loss* penalises violations of the heat equation at a set of N_f interior collocation points $\{(x_i^{(f)}, t_i^{(f)})\}$:

$$\mathcal{L}_{\mathrm{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \left| R_\theta(x_i^{(f)}, t_i^{(f)}) \right|^2.$$

The *initial-condition loss* enforces the prescribed initial temperature profile $f(x)$ at N_0 spatial points sampled at $t = 0$:

$$\mathcal{L}_{\mathrm{IC}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \left| T_\theta(x_i^{(0)}, 0) - f(x_i^{(0)}) \right|^2.$$

The *boundary-condition loss* enforces the prescribed boundary temperatures g_0 and g_L at both ends of the domain, sampled at N_b time points:

$$\mathcal{L}_{\mathrm{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| T_\theta(0, t_i^{(b)}) - g_0(t_i^{(b)}) \right|^2 + \frac{1}{N_b} \sum_{i=1}^{N_b} \left| T_\theta(L, t_i^{(b)}) - g_L(t_i^{(b)}) \right|^2.$$

These three terms are combined into the total loss

$$\mathcal{L} = \lambda_{\mathrm{PDE}} \, \mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{IC}} \, \mathcal{L}_{\mathrm{IC}} + \lambda_{\mathrm{BC}} \, \mathcal{L}_{\mathrm{BC}}.$$

The integers N_f, N_0, and N_b control how densely each constraint is sampled: N_f collocation points are drawn from the interior of the space-time domain to enforce the differential equation, N_0 points along the initial-time slice enforce the initial condition, and N_b points along each boundary enforce the boundary conditions. The scalar coefficients λ_{PDE}, λ_{IC}, and λ_{BC} are user-chosen hyperparameters that balance the relative importance of the three constraints. Increasing a particular λ directs the optimiser to prioritise satisfying that constraint more strongly. The network is trained by computing gradients of $\mathcal{L}$ with respect to all weights and biases θ and updating them iteratively using gradient descent, until $\mathcal{L}$ is acceptably small.

Summary of the workflow

In brief, solving the 1D heat equation with a PINN proceeds as follows:

(i) Choose a space-time point (x, t) in the domain.

(ii) Evaluate the network to obtain the predicted temperature $T_\theta(x, t)$.

(iii) Compute the partial derivatives $\partial T_\theta / \partial t$ and $\partial^2 T_\theta / \partial x^2$ via automatic differentiation.

(iv) Form the PDE residual $R_\theta(x, t)$ by substituting these derivatives into the heat equation.

(v) Repeat steps (i)–(iv) over many sampled points and evaluate the total loss $\mathcal{L}$ over all collocation, initial, and boundary sets.

(vi) Update the network parameters θ by back-propagating gradients of $\mathcal{L}$, and iterate until convergence.

At the end of training, $T_\theta(x, t)$ is a continuous, differentiable approximation of the solution that satisfies the heat equation, initial condition, and boundary conditions simultaneously, without any spatial grid or time-stepping scheme.

2 Experimental Setup

A setup of our system is shown below in Figure 1.

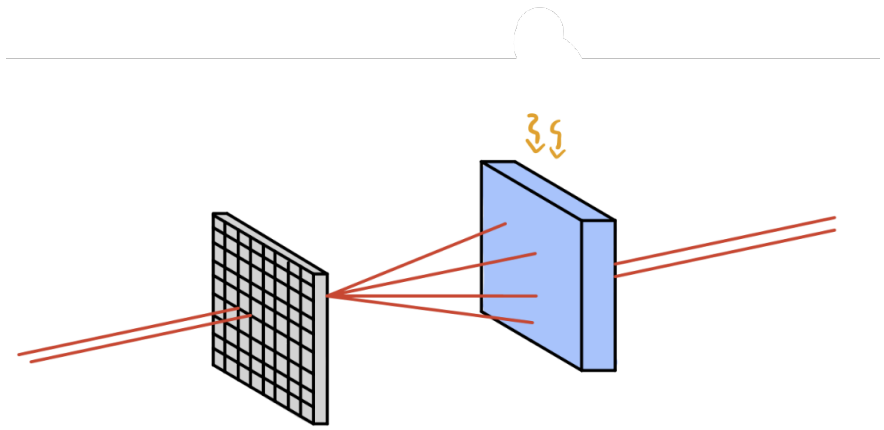

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the thermo-optical material, and is detected by the camera.

The optical process is summarized as follows:

i. Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

ii. Apply SLM phase modulation to obtain $E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}$.

iii. Propagate through the thermo-optical material using the transmission matrix $T_{\text{opt}}(t)$ to give $E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}$.

iv. Compute the output intensity $I(x, y) = |E_{\text{out}}(x, y)|^2$.

v. Define the reinforcement-learning reward as the intensity at a selected target output location.